
“Value Creation and Value Capture”

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I. What is value?

The questions, “What is ‘value’?” and “What do humans value?” are both incredibly important and extremely complex. Numerous fields of study and intellectual perspectives provide different lenses for thinking about these questions. Even religious documents provide perspectives, including:

“Do not toil to acquire wealth; be wise enough to desist. When you set your eyes on it, it is gone, for it will surely sprout wings and fly away like an eagle.” - Proverbs 23:4-5

Philosophers provide other perspectives, including:

“Of course, the central question philosophers have been interested in, is that of what is of intrinsic value, which is taken to contrast with instrumental value. Paradigmatically, money is supposed to be good, but not intrinsically good: it is supposed to be good because it leads to other good things: HD TV’s and houses in desirable school districts and vanilla lattes, for example. These things, in turn, may only be good for what they lead to: exciting NFL Sundays and adequate educations and caffeine highs, for example. And those things, in turn, may be good only for what they lead to, but eventually, it is argued, something must be good, and not just for what it leads to. Such things are said to be intrinsically good.” – Stanford Encyclopedia of Philosophy, “Value Theory,” Section 2.1 “Intrinsic Value” ([link](#))

In this course, we will not argue that there is a universal definition of value or that value is inherent in any particular concept or object:

“Value is thus nothing inherent in goods, no property of them, nor an independent thing existing by itself. It is a judgment economizing [people] make about the importance of the goods at their disposal for the maintenance of their lives and well-being. Hence value does not exist outside the consciousness of [the mind].” — Carl Menger, 1871

We acknowledge that different individuals, different cultural traditions, and different approaches to thought define value differently and hold varying values. For example, moral and ethical values, like the extent to which a society values truth and justice, vary over time, place, and person.

We acknowledge, as well, that moral and ethical values are foundational for economic and business activity. Over the course of the semester, the course will introduce frameworks for thinking about morality and ethics in business and we will apply this in ways that recognize individual differences in values.

We will, however, introduce a particular and powerful framework for thinking about **economic value**. As we do, we keep in mind that economic value may not reflect the entirety of moral and ethical values and that the differences between these may have a



Source: Routledge Blog, “Nature: Can we put a value on it?” (2022)

powerful impact on the ways businesses behave.

II. What is economic value?

II.1 Labor as a source of economic value

Economists have wrestled with the question of, “what is economic value?” since at least the time of **Adam Smith** (1723-1790). In his famous book, *An Inquiry into the Nature and Causes of the Wealth of Nations*, which was published in 1776 and is commonly referred to simply as, *The Wealth of Nations*, Smith articulated a view that came to be known as, “the labor theory of value.” This view contended that the value of goods and services was determined by the amount of labor required for their production:¹

“Labour is the real measure of the exchangeable value of all commodities. The real price of everything, what everything really costs to the man who wants to acquire it, is the toil and trouble of acquiring it. What everything is really worth to the man who has acquired it, and who wants to dispose of it or exchange it for something else, is the toil and trouble which it can save to himself, and which it can impose upon other people. What is bought with money or with goods is purchased by labour as much as what we acquire by the toil of our own body.” (*Wealth of Nations*, Chapter 4)

David Ricardo (1772-1823), who would eventually introduce the theory of comparative advantage, agreed with Smith on the role of labor in driving value, as did a number of other famous economists and natural philosophers of the 18th and 19th centuries.

Karl Marx was perhaps the most well-known adherent of the labor theory of value. In his classic work, *Das Kapital* (1867), Marx makes the claim that the value of a product can be measured objectively based on the average number of hours of human labor required for its production (CITE). By his logic, if it were to take twice as long for an individual to produce a shirt as to produce a hat, then shirts would be twice as valuable as hats, i.e., because they required twice as much human labor.

Smith wrote at the time that the Industrial Revolution was beginning to take root. He famously observed early factories and noted that productivity gains were the result of early machines and the specialization of labor, i.e., having workers become expert in steps in the production process rather than having workers fully assemble products from start to finish. At the time Smith wrote the *Wealth of Nations*, steam-power machines had not yet become ubiquitous in complementing and substituting for human labor.² By the time Marx was writing in the mid 1800s, however, industrial

¹ Smith links labor and value in numerous additional text in his work, including the following two passages, each of which appears in Chapter 5: “The value of any commodity, therefore, to the person who possesses it, and who means not to use or consume it himself, but to exchange it for other commodities, is equal to the quantity of labor which it enables him to purchase or command. Labor, therefore, is the real measure of the exchangeable value of all commodities.” “Labor alone, therefore, never varying in its own value, is alone the ultimate and real standard by which the value of all commodities can at all times and places be estimated and compared. It is their real price; money is their nominal price.”

² Things changed quite dramatically over the next few decades following the publication of the *Wealth of Nations*. By the 1810s textile machinery had begun to displace so many textile workers that a protest movement was formed to resist the introduction of labor-saving textile machines. Named after an apocryphal individual, Ned Ludd, the “Luddite” movement threatened mill owners and destroyed textile machinery machines in the effort to oppose industrialization and preserve textile jobs. Originally referring to textile workers fighting against job-replacing machines, “Luddite” now applies to anyone fearful of or opposed to technological advancements.

urban factories had begun to dominate business in England and the role of machines in creating economic value could have been observed more readily.

II.2 Economic value as objective (or intrinsic)

The labor theory of value is one of various ways of conceiving of economic value as theoretically objective (or intrinsic). In this approach to thinking of economic value, items have a value that is fully determined by their **objective features**. In this view, things are not valued based on human perception or preferences but because of their inherent importance or as a result of the physical or human components of their production.

William Petty (1623-1687) and **Richard Cantillon** (1680-1734) are two other writers who espoused views suggesting that economic value could be objectively determined. Petty argued that the natural value of products would reflect the amount of land and labor required for their production, while Cantillon argued that the value of a good depended on land alone. These basis for these authors' views and for the views of Adam Smith that we introduced earlier were, in part, based on the fact that contemporaneous methods of production were often relatively simple. When most products are agricultural or are produced without substantial equipment, one can imagine why authors considered the drivers of value to be relatively straightforward. As technology advanced and the scale and complexity of production grew, ideas about economic value evolved to include the idea that economic value was not objective or intrinsic.

While Smith wrote repeatedly in *The Wealth of Nations* of the role of labor in determining the value of simple goods, he suggested an expanded (though also objective) approach to measuring economic value in cases in which production was more complex. He noted that, in many production processes of the time, workers manufactured products utilizing machinery to convert raw materials into finished products. Further, many such products required substantial land usage or were the results of products produced based on substantial land usage. In such complex production processes, Smith argued that value should be understood via a process of **"adding up."**³ By this approach, Smith suggested that the value of a complex product would result from the sum of labor costs, the (measurable) profit of the capitalist, and the rent associated with the land used to make the product.

As Smith's approach to thinking about the economic value of complex goods suggests, the idea that economic value could be understood as objective was related to simplicity of early methods of production. As technology advanced and the scale and complexity of production grew, ideas about economic value evolved to include the idea that economic value was not objective or intrinsic.

2.3 Subjective approaches to Economic Value

Smith, Ricardo, and Marx are often considered to be representatives of the "classical" school of economics. At around the same time that Marx was writing *Das Kapital*, a set of other economics were formulating an approach that is more consistent with the approach taken by modern economics.

³ "In the price of commodities, therefore, the profits of stock constitute a component part altogether different from the wages of labour, and regulated by quite different principles ... As soon as the land of any country has all become private property, the landlords, like all other men, love to reap where they never sowed, and demand a rent even for its natural produce. The wood of the forest, the grass of the field, and all the natural fruits of the earth, which, when land was common, cost the labourer only the trouble of gathering them, come, even to him, to have an additional price fixed upon them. He must then pay for the licence to gather them; and must give up to the landlord a portion of what his labour either collects or produces. This portion, or, what comes to the same thing, the price of this portion, constitutes the rent of land, and in the price of the greater part of commodities makes a third component part" (*The Wealth of Nations*, Ch 6).

One of the thinkers whose ideas helped progress past the objective approach was **John Stuart Mill** (1806-1883). In his 1838 book, *The Principles of Political Economy*, Mill wrote, “The value which a commodity will bring in any market is no other than the value which, in that market, gives a demand just sufficient to carry off the existing supply” (Chapter 2).

The idea underlying this **subjective approach** to measuring economic value is that the value of goods and services is not inherent, intrinsic, or objective, but is, instead, shaped by a combination of the usefulness of a product or service to its potential buyers and by the preferences of those buyers. In this approach, which we introduced in this note in the opening quote by Menger, the value of goods and services can vary from place to place, year to year, or even minute to minute.

2.4 Economic Value and the “Marginal Revolution”

A series of economic theorists helped develop the modern approach to understanding economic value, which considers economic value. These included **Léon Walras** (1834–1910) and **William Stanley Jevons** (1835–1882). Walras, Jevons, and others initiated the “marginal revolution” in economics. This revolution began to employ the tools of mathematics, particularly the calculus invented by Sir Isaac Newton and Gottlieb Leibniz, in order to develop abstract, formal theories built using mathematical tools, in which economic value was determined subjectively.

The ideas of the ‘marginalists’ coalesced around an understanding of economic value based on the idea of utility, i.e., the usefulness of a product or service to whomever might use it. Unlike the approach in which value was driven by the inherent nature of the product or service or the amount of labor required to produce it, utility could vary across individuals, groups, and over time.⁴ In this conception, value is subjective, though still measurable. In particular, the value of a good or service could be measured based on the amount that potential buyers were willing to pay for it. Thus, if a society decided that shirts were ten times as important as hats and people were willing to pay ten times as much for a shirt as for a hat, then the value of the average shirt would be equivalent to the value of ten average hats.

Some aspects of utility are quite functional. For example, farmers derive utility from harvesting equipment based on how efficiently it can harvest crops. Some aspects of utility may be more subjective. For example, farmers may derive additional utility from harvesting equipment because of its visual appeal and ease-of-use, even if those features do not enhance its productivity. Overall, modern economics and business recognize that the economic value of goods and services is not directly determined by innate features of products or the number of labor hours that went into making them but by the complex interplay of social, economic, and moral processes that lead individual or business consumers to derive utility from those goods and services.

Though utility forms the basis for modern economists’ notions of value, utility itself is hard to observe. What can be observed in the behavior of human and business consumers, however, are the

⁴ Microeconomics 101 highlights another important notion of the ‘marginalists,’ which is the idea that the utility of a good or service typically decreases as a function of the amount consumed. Someone who loves cookies may derive substantial utility (happiness) from eating a cookie. However, the more cookies they consume, the lower will be the additional value of each additional cookie. For example, the tenth cookie that the person consumes may not bring as much joy as the first. The ‘marginalists’ refer to the additional value derived from each extra unit of consumption as the “marginal value.” Those who have completed calculus may recognize that the marginal value can be represented mathematically as the slope of the utility function when Utility is a function of the Quantity Consumed.

prices that individuals and businesses are willing to pay for goods and services.⁵ As we will discuss in greater detail later, economists will therefore consider the average economic value of a good or service to be equivalent to the average price that a buyer would pay for it.

III. What is a Value Chain?

We are nearly ready to introduce the value creation and value capture framework. Before we do this, though, we have one more idea to introduce. This idea is that of a simple **value chain**. The simple value chain describes the path of exchange of goods and services (a) from a supplier to a firm and (b) from that firm to a buyer. The **simplified value chain** in this scenario looks like this:

Figure 1: Value Chain (single supplier, single firm, single buyer)



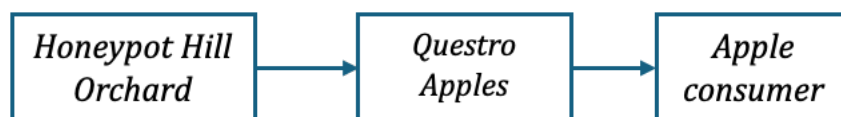
Note: In a value chain, goods and services flow from left to right (from supplier to the firm to buyer), while payment flows from right to left (from the buyer to the firm to the supplier).

We will introduce more complex value chain diagrams in the future. Such value chains will involve more steps in the production process, i.e., a longer chain of suppliers and buyers.

In all of our value chain diagrams, goods and services will flow from left to right (e.g., from supplier to firm to buyer), while payment for those goods and services flow from right to left (e.g., the buyer pays the firm and the firm pays its supplier). The most complex value chain diagrams will begin on the left-hand side with raw materials and will end on the right-hand side with purchases by end consumers (or the users for whom the consumers purchase the goods and services).

Let's consider a simple example of QuestroApples, a fictional apple stand that sells nothing but apples. If QuestroApples were to purchase its apples from Honeypot Hill Orchard (HHO), we would consider HHO to be its supplier. If it sold its products to an apple consumer, we would consider the apple consumer to be the buyer.

Figure 2: Value Chain: QuestroApples example



Note: In a value chain, goods and services flow from left to right (from supplier to the firm to buyer), while payment flows from right to left (from the buyer to the firm to the supplier).

We typically draw value chains on a horizontal axis, i.e., writing them from left to right. When we consider use value chains to consider value creation and value capture, as we will do in the next step, we can also represent them vertically, as this will help in understanding the process of value creation and capture.

IV. The Value Stick – a framework for thinking about Value Creation and Value Capture

4.1 Value Creation – the two-party scenario

To structure our thinking about Value Creation, let's start with a simple scenario that considers a value chain that includes only a single step, a step in which a Firm wants to sell its product to a

⁵ Economists refer to these behaviors as demonstrating consumers' **revealed preferences**. The idea is that we may never be able to get individuals to disclose their internal preferences – indeed, folks may not know themselves what their underlying preferences are – but we can deduce preferences based on individuals' behaviors.

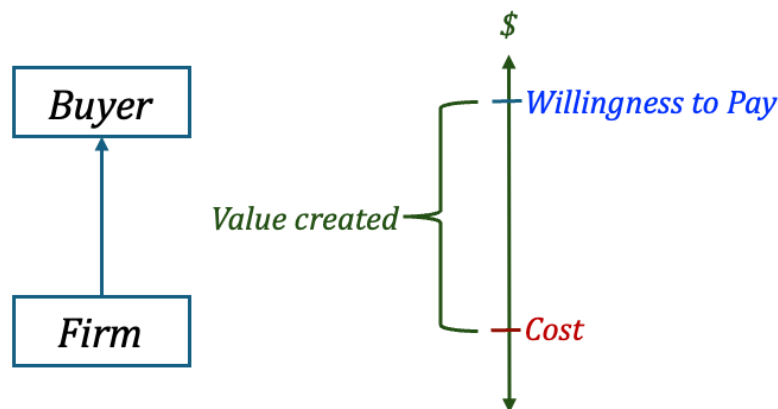
Buyer. In this scenario, value creation is defined based on only two components: (a) the **willingness to pay** (WTP) of the buyer and (b) the **cost** to the firm. Thus, the value created in this step can be defined as follows:

Formula 1: **Value Created = Willingness-To-Pay – Cost**

This equation represents the value created by the firm. The cost reflects the average amount that the firm pays for supplies and manufacturing per unit of product it sells, while the willingness to pay reflects the average amount that buyers of the firm's product would pay to obtain the product. Value created thus reflects the investments of the firm in transforming its supplies into a product (or service) into something that its buyers would want.

Figure 3 depicts this relationship in graphical form, using something we can refer to as the **Value Stick**. The Value Stick is similar to the Value Chain diagram we introduced before, but it is stacked vertically rather than horizontally in order to make it easier to represent value creation and capture. The figure below includes only the value creation component, but we will expand the stick in a moment to reflect value capture as well.

Figure 3: Value Creation – the two-party scenario



Note: In this figure, Willingness to Pay (WTP) reflects the average amount of dollars that the average consumer would be willing to pay for the firm's product (or service), while the Cost reflects the average cost to the firm of producing a unit of its product.

The formula for value creation seems straightforward, but it is both powerful and more complex than one might imagine. Notice that the amount of money that the buyer actually has to spend to purchase the product does not play any role in the equation. That's right: the amount of value created is not determined by the product's price. Instead, the average value created in any single step of the value chain depends just on the average cost to the firm and the average willingness to pay of its buyers.

We will discuss both Cost and Willingness to Pay in a bit more detail in Section V, which explores how firms create value, but we should say a few words about each here. WTP is a nuanced concept. It reflects the hypothetical maximum amount that a Buyer would spend for a good or service, assuming that they have the resources to make the purchase. If the price for a good or service were above a Buyer's Willingness to Pay, the Buyer would not buy it, regardless of their ability to pay for it. If the price is Willingness to Pay, they would make the purchase.

Willingness to pay reflects the utility of a product or service to its potential buyer. WTP can depend on both (a) the productive utility of a product or service and (b) features of the product or service

that appeal to the buyer. For example, willingness to pay for a shampoo might be based on both the utility of the shampoo in serving its function of cleaning and supporting healthy hair and on other features that have appeal to the buyer other than its pure functional utility, such as the buyers' perception of the shampoo's brand, packaging, and smell. Business purchases of industrial equipment are typically more closely related to the functional utility of products, like the harvester equipment we described above, while purchases of consumer products, like shampoo and fashion products, are more likely to incorporate a significant component related to buyers' impressions of those products. It is also important to note that willingness to pay varies significantly across consumers. When thinking about value creation, we should focus on the average amount that buyers would be willing to pay for a product.

Average Cost is a concept that is a bit less nuanced than Average WTP, as we will see in our examples. From the perspective of the firm that makes the product or service, Cost reflects the average amount that the firm spends on all things required to make the product or service available for sale. One way to approximate average cost is as the firm's total costs, i.e., the costs of all firm expenditures in a particular unit of time (e.g., in a year or over five years), divided by the total amount of units sold in that time period.

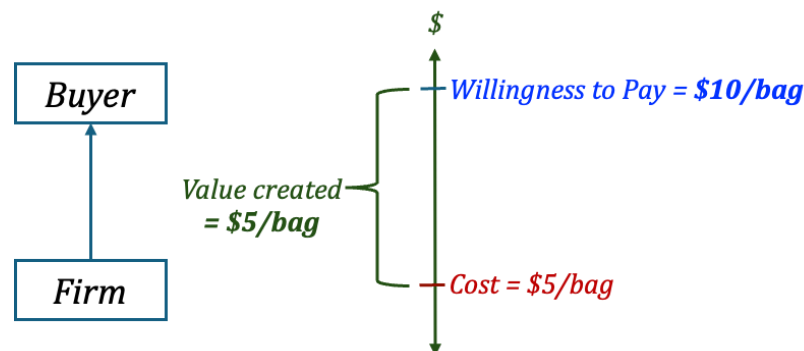
4.2 Value Creation – an example using the two-party scenario

Let's return to our example of QuestroApples. Suppose QuestroApples were able to obtain its apples at a cost of \$5 per bag and the average buyer were willing to pay \$10 per bag.

In this scenario, we would consider QuestroApples as creating \$5 of value, as reflected in the formula and figure below:

$$\text{Value Creation } (\$5) = \text{Willingness-to-Pay } (\$10) - \text{Cost } (\$5)$$

Figure 4: Value Creation – two-party example (QuestroApples)



4.3 Value Capture in the two-party scenario

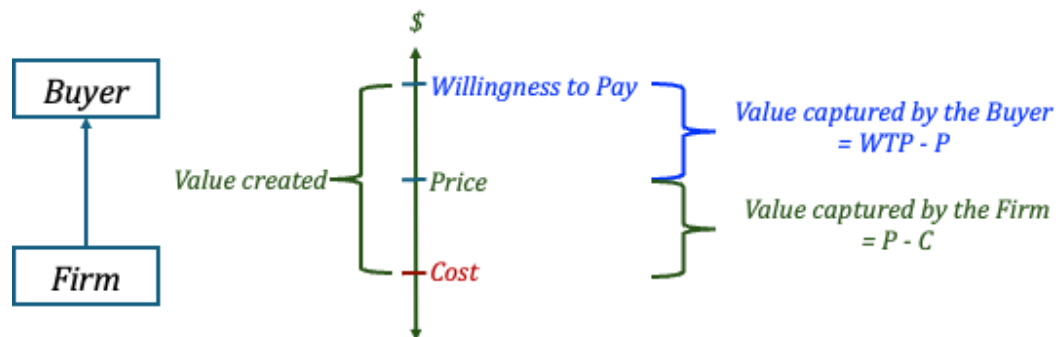
Now that we have explained value creation, we can add the exciting step of understanding **value capture**. When we thought about value creation in our simplified value chain, we only focused on one party, the firm, because only the firm was engaged in an activity that transformed supplies into products or services. In the simplified value chain we considered, all that the buyer did was consume the product. In more complex value chains, in which buyers in one stage may be sellers in another stage (for example, when a beverage maker sells drinks to a retailer, which then sells those

drinks to an end consumer), we will examine the value created by each production step in the value chain.

However, even in the simplified value chain in which there are two parties, we need to be clear about which one we are considering when we think of value capture. We can refer to this as the **division of value** in the value chain.

Price turns out to be the key concept that helps us understand value capture. In a simplified value chain with only a buyer and a firm (i.e., the firm that is selling the product), the price clarifies how much value is captured by the buyer and how much is captured by the firm. This is depicted in Figure 5= below.

Figure 5: Value Capture – two-party scenario



Thus, the Value Captured by the Firm and Buyer, respectively, are described by the formulae:

Formula 2: **Value Captured by the Firm (also known as Profit) = $P - C$**

Formula 3: **Value Captured by the Buyer = $WTP - P$**

These are quite intuitive if you think about them. Firm profits reflect the difference between average prices and average cost. Thus, the Value Captured by the Firm, also known as Firm Profit, is equal to the average price the firm receives for each product sold and the average cost of that product. The Value Captured by the Buyer is described as the difference between what the Buyer would have been willing to pay for the product and the price that they actually pay. This is known in economics as “Consumer Surplus” and it is sometimes referred to in business writing as “Consumer Delight.”

You may be able to think about some examples of cases in which your own purchases involved prices that were close to or far away from the maximum that you would have been willing to pay. Typically, consumers are quite happy when they are able to buy a product for much less than they would have been willing to pay.

These formulae highlight a number of interesting facts: In a two-party value chain, value is can be captured by either party. The Buyer’s Willingness-to-Pay and the Cost to the Firm are the key concepts in understanding Value Creation. Price is important in understanding the division of value between the Buyer and the Firm.

For the moment, do not worry about whether all of the products that the firm makes gets sold or whether some sell for different prices or whether the firm makes multiple types of products. For conceptual purposes, think simply about the average price and average cost across the entire

portfolio of the firm's products. For example, BMW sells many types of automotive vehicles at many different prices, but we could identify an average price and average cost for a BMW product and get a sense for the average value BMW creates and captures.

4.4 Value Capture – an example using the two-party scenario

To illustrate the value capture framework, let's return to our QuestroApples example. Let's suppose that QuestroApples sold their apples at an average of \$8 per bag and that Buyer WTP and the Cost per bag remained at \$10 and \$5 per bag, respectively.

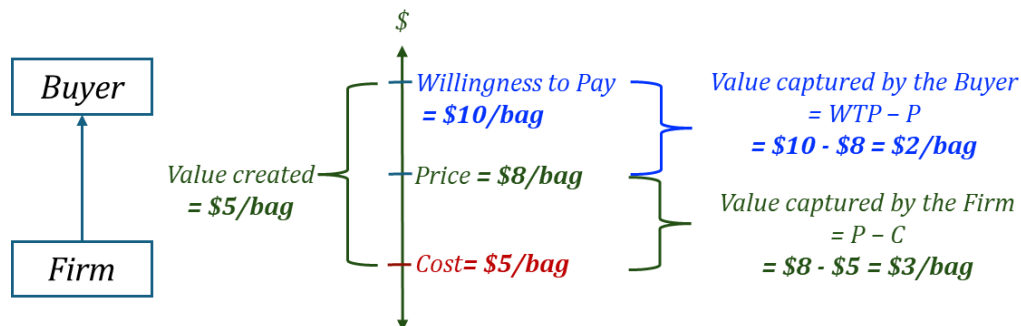
In this scenario, which is depicted in Figure 6 below, Value Created, Value Captured by the Buyer, and Value Captured by the Firm would be computed as follows:

Value Created would remain at \$5 (Value Created = $WTP - C$, i.e., $\$10 - \$5 = \$5$).

Value Captured by the Buyer (aka, Consumer Surplus) would be \$2 per bag (Value Captured by the Buyer = $WTP - P$, i.e., $\$10 - \$8 = \$2$).

Value Captured by the Firm (aka, Profit) would be \$3 per bag (Value Captured by the Firm = $P - C$, i.e., $\$8 - \$5 = \$3$).

Figure 6: Value Creation & Capture – two-party example (QuestroApples)



The division, of value between the Buyer and the Firm would change if prices changed. To some extent, firms have control over the prices they charge. But, as economics shows, prices are also driven by microeconomic conditions, i.e., supply and demand. If, for example, a jump in demand might enable prices to rise, thus enabling firms to capture more value and for consumer surplus to decline. If, by contrast, the market became more competitive, possibly because a rival apple stand

One final note is that all of the value in the two-party value chain is created by the firm, during the one step in the value chain, the step in which QuestroApples sells apples. In the section that follows, we will unpack the way in which firms like QuestroApples can create value.

V. How can firms create value?

5.1 Overview

Now that we have formulas and a framework to think about the creation and capture of economic value, we can ask this question: How can firms create value? The frameworks suggest that the two categories of things that firms can do to create value are either (1) to increase average willingness to pay or (2) to decrease average costs.

Understanding how to do this requires a bit more explanation of what willingness-to-pay and average cost are and of the factors that drive both.

5.2 Drivers of Willingness to Pay

As we noted in Section 4.1, average willingness to pay reflects both the productive utility of a product or service to the buyer and features of the product or service that appeal to the buyer. It is conceptually possible to separate productive utility from appeal. In the example we used above, we noted that the productive utility of harvesting equipment to a farmer will depend on how efficiently it is able to contribute to harvesting. The color of the harvesting equipment may not affect its productivity but could, theoretically, affect the appeal of the equipment to the farmer. Product features like the comfort of using the harvesting equipment could affect both its efficiency and its appeal in ways that do not affect its efficiency.

Investments that improve the functional utility of a good or service are likely to increase WTP. For example, investments that enable dishwashing detergent to clean more dishes while using the same amount of detergent would likely increase WTP for detergent, investments that reduce the side effects associated with a medical treatment while providing the same health benefits, or investments that enable a semiconductor to increase the number and speed of its calculations, are all likely to increase WTP.

A firm may also be able to increase WTP for a product or service by developing a new variety of that product that satisfy previously unaddressed customer needs. For example, by designing soccer cleats with adjustable-length spikes, which enabled the shoes to adapt to multiple field conditions, Adi Dassler, founder of Adidas, increased willingness to pay for soccer shoes in the 1950s.⁶

This example highlights the role of **innovation** as a path to increase WTP. Such innovations could take the form of new products and services, such as Adidas's new cleats, or improvements in existing products and services. For example, although headphones had existed for decades, the Boston-area firm Bose increased WTP for a new category of headphone when it developed and commercialized the world's first commercial active noise-canceling headphones in 1979. Innovations that increase WTP can include **product innovations**, **process innovations** (innovations in the way in which products or services are made), or **technological innovations** that enable the development or enhance the features of new products and services.

Other investments or choices that firms can make that affect the usefulness of a product and service include making it available to consumers in a way that makes it easier for them to obtain it or to feel comfortable using it. In our earlier example of the apple stand, some of the value that QuestroApples provides may derive from its location, i.e., ensuring that consumers are able to

⁶ Dassler's invention is also credited with contributing to Germany's first-ever FIFA World Cup championship in the FIFA World Cup back in 1954. Although Hungary entered the final unbeaten in its previous thirty matches, only the German national team had access to Dassler's adjustable-length cleats, which appeared to give the team an advantage as they overcame an early 2-0 deficit and rainy conditions to come back and upset Hungary 3-2 in what is known soccer history as, "The Miracle of Bern."

obtain apples without having to make a special trip to a distant location. QuestroApples may also develop a reputation for quality that contributes to WTP. If QuestroApples develops a reputation for selling apples that are fresh, tasty, and free of harmful chemicals, this reputation may contribute to the product's utility, thus boosting WTP.

In addition to investments that increase WTP by improving product utility, firms can make **investments in the subjective product features that do not affect a product's usefulness but do affect its appeal to Buyers**. For example, if QuestroApples introduced a visually appealing logo produced a catchy jingle that were unconnected with the quality of its products, this might boost WTP for its apples.

As we noted in Section IV, the value of subjective features unconnected with product utility is more likely to affect WTP for consumer products than industrial products, though it can arise in both contexts.

Figure 7 highlights some of the factors that may affect WTP, though it does not present an exhaustive list. The figure highlights some factors related to the productive utility of products and services, as well as some factors more closely associated with a product's subjective features that may not be associated with its quality or usefulness.

Figure 7 - Factors that can impact Willingness to Pay[^]

• income	• geography
• weather	• age
• gender	• brand loyalty
• service levels	• advertising
• competing products	• expectations
• legality	• packaging
• environmental or social impact	• necessity

[^] Note that this table replicates the content of a graphic in Stobierski (2020), though with simplified formatting.

5.3 Drivers of Cost

As we noted above, average cost is more straightforward to understand than WTP. One straightforward way to think about the average cost is by considering the total expenditures that a firm bears in producing all of the goods and services it is able to sell in a particular time period and by dividing that by the total amount of units it sells during that time period. This will yield an approximation of the firm's average cost. It averages out across multiple product varieties and it embeds a number of assumptions about how big-ticket costs (known as fixed costs) are allocated over time. Getting these numbers exactly right is complex, but approximations are useful, even though it is hard to get all the way to the perfect estimate.

For example, in the case of QuestroApples, we would need to consider the total amount that QuestroApples expends to be able to sell its apples. This would include the costs associated with obtaining apples for it to sell, the costs associated with buying the bags in which it packaged its apples, the costs of labor and associated human resource benefits, of the physical apple stand, of the rent required to use the land on which the apple stand was located, of the fire and other insurance required to operate the stand, etc.

We noted above that firms can create value by increasing Average Willingness to Pay via investments that improve the functional utility or perceptions of product features. Innovations often play a key role creating value by boosting WTP. Similarly, firms can create value by engaging in investments, including innovation-oriented investments that drive down average costs. Innovations that enable a firm to produce more goods and services at the same level of quality or by using less human capital or physical capital would each qualify as cost-reducing, and thus value creating, innovations.

For example, if QuestroApples were able to rearrange its apple stands in order to sell the same number of apples using fewer workers, this would decrease average cost and create additional value.

VI. How can firms capture value?

How can firms capture value? This is an important, indeed, an essential question for businesses. If all of the value they create is captured by Buyers, they may not be able to continue operation.

As the framework notes, the ability of a firm to capture value is different from the ability of the firm to create value. The reason for this is that while firms can make choices that affect WTP and Cost, only those choices that drive down Cost (while preserving WTP) are certain to lead to Value Capture.

The reason for this is that not all WTP increases are accompanied by increases in Price. If a firm makes a series of investments that increase WTP but do not yield an associated increase in Average Price, then the Value Captured by the WTP increase will redound to the Buyer, as Consumer Surplus, rather than as increase Profitability.

If an increase in WTP leads to greater volume of sales, which, in turn, drives down Average Costs, this can enable Value Capture as well. As a purely conceptual matter, though, it is important to remember that increases in WTP are associated with increased Value Creation, though not necessarily increased Value Capture by the Firm.

These are the main points that we would like to make about Firm Value Capture for the moment, but note that we will dedicate substantial additional time to thinking about how firms can capture value as the course develops.

VII. Extending the Value Creation and Value Capture Framework to >2 Parties

7.1 Value Creation and Value Capture in an extended value chain

So far, we have constrained our analysis to a simple, two-party value chain. Most value chains, however, involve more than one transaction. We can extend the value chain either upstream, i.e., closer to the raw materials, or downstream, i.e., closer to the end consumer, depending on which additional steps in the value chain exist. If the Firm in our two-party case obtains all of its inputs from a single Supplier, we could add that Supplier to the bottom of our graphic, as in Figure 8

Figure 8 – extending the Value Chain to additional parties

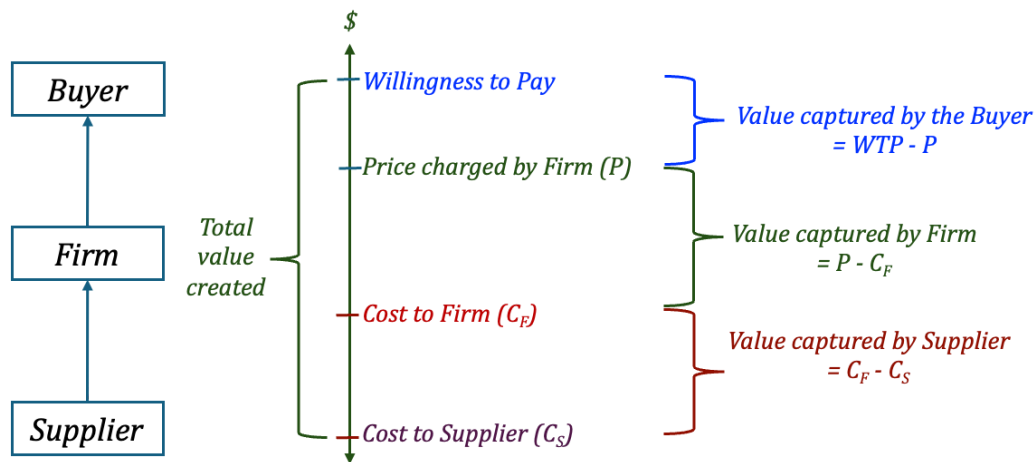


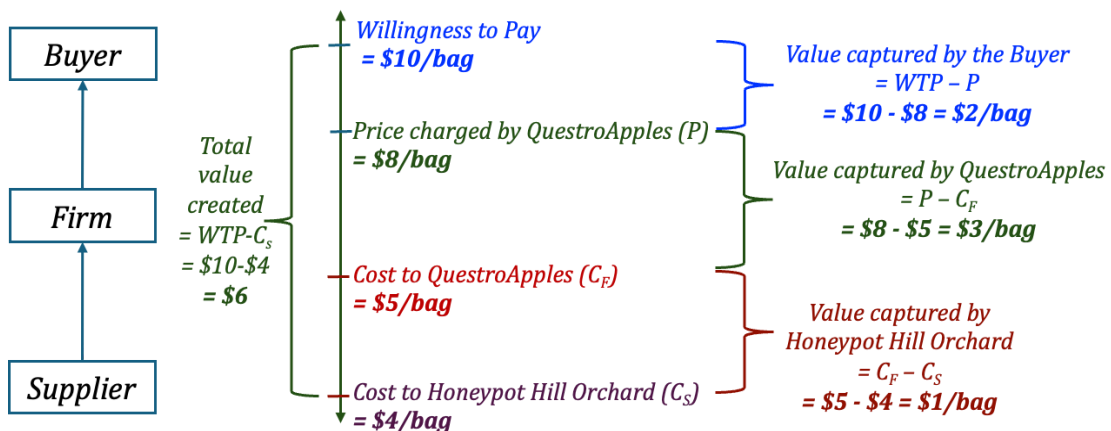
Figure 8 is similar to Figure 5, although the addition of the Supplier allows us to generalize Value Creation and Value Capture across the value chain.

The Figure shines light on a few features of the generalized value chain. First, highlights that Total Value Created in the value chain is equal to the End Consumer's Willingness to Pay minus the average cost of inputs to the firm at the beginning of the chain. Second, it allows us to see that the Value Captured by each firm in the value chain can be computed as the Average Price at which that firm it sells its products minus the Average Cost it bears. For the Firm on which we are focusing in the generalized value chain, its Average Cost reflects the Price at which its Supplier sells its products. From the perspective of the Supplier, therefore, Value Captured can be computed as the difference between the Average Cost to the Firm and the Average Cost to the Supplier.

7.2. The extended value chain using a now-familiar example

When we introduced the example of QuestroApples in the two-party value chain, we noted that the firm purchased its apples from Honeyplot Hill Orchard (HHO). Figure 9 below relates what the extended value chain would look like if we were to also consider the value created and captured by HHO, assuming that the average cost of all inputs to HHO were \$4 per bag.

Figure 9 – Value Creation & Capture – example using the extended Value Chain



If this were the case, then the following relationships would hold:

Value Created across the extended value chain would be \$6 (Value Created = $WTP - C_S$, i.e., $\$10 - \$4 = \$6$).

Value Captured by the Buyer (aka, Consumer Surplus) would remain \$2 per bag (Value Captured by the Buyer = $WTP - P$, i.e., $\$10 - \$8 = \$2$).

Value Captured by the Firm (the Firm's Profit) would remain \$3 per bag (Value Captured by the Firm = $P - C_F$, i.e., $\$8 - \$5 = \$3$).

Value Captured by the Supplier (the Supplier's Profit) would be \$1 per bag (Value Captured by the Supplier = $C_F - C_S = \$5 - \$4 = \$1$).

7.3. Additional extensions we do not pursue here

There are a number of other extensions of the value creation and capture framework that we could pursue and should mention, but which is outside the scope of the current note.

One extension is that we could include even more upstream suppliers. This extension would simply generalize the three-party case by adding additional suppliers in exactly the same manner and with exactly the same implications as the addition of the third party, i.e., the Supplier.

Another extension we could pursue would involve a more complete exploration of the impact of rivalry among Firms. This would stretch the Value Creation and Capture framework a bit, but the main implications are that competition via rivalry among firms in similar positions in the value chain is likely (1) to have a direct effect on the prices at which rival firms sell their goods and (2) to have an indirect effect on willingness-to-pay and costs along the value chain as rival firms compete on product and process innovation and apply pressure to their suppliers to contribute to WTP-enhancing innovation and to lower costs.

VIII. Conclusions and Takeaways

This note has introduced a number of concepts of relevance for business and management. First, we have introduced the idea of **value** and noted that **economic value** is a particular conception of value. Second, we outlined some of the **history of ideas regarding economic value**, emphasizing the evolution of these ideas from objective, intrinsic conceptions of economic value to subjective conceptions based on the ideas proposed during the Marginal Revolution. Third, we introduced a **framework** for thinking about **Value Creation and Value Capture** that builds on a simple two-party **Value Chain** and we extended into a multi-party value chain.

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